Results and their relation to prior open problems

Critical focusing dispersive systems: thresholds, coherent structures and dynamics | 8 October 2026 | Rounds 1–13

Scope. This inventory includes the early computer-assisted counterexamples, their later extensions, and the subsequent analytic theory. It corrects the recent conversational omission of the early results. It is a status synthesis, not a new proof or an external referee report.

1. The original universal collapse assertion has a negative answer

The four-dimensional mass-critical equation has exact nonradial traveling solutions with negative energy and a constant finite H^2 norm for every real time. The later three-dimensional construction gives the same conclusion for its own mass-critical power. Thus negative energy alone does not force nonradial collapse in either dimension.

$$i\,u_t = \Delta^2 u - |u|^{(8/N)}u;\ N = 4 \text{ or } N = 3\ u(t,x) = \exp(it)\,U_v(x - vt\,e_1),\ E(U_v) < 0.$$

Boulenger–Lenzmann [1], §1.5, explicitly discuss removing radial symmetry. These examples give a negative answer to the unconditional negative-energy extension in that direction. They do not refute the paper’s radial theorems or its separate radial finite-time conjecture, and do not settle every fixed dimension.

Proof location: BNLS_Nonradial_Research_20261007, Theorem 1.1; BNLS_Zero_Momentum_and_Three_Dimensions_20261007, Theorem 1.2.

2. A direct counterexample to one precise prior theorem

Gou [2], publicly readable arXiv v3, Theorem 1.2, assumes cylindrical symmetry, H^2 regularity, a finite axial L^2 moment and negative energy at mass criticality. It concludes finite-time blowup for $\mu > 0$ and finite- or infinite-time norm growth for $\mu = 0$. The four-dimensional profiles satisfy those assumptions and remain globally bounded. Continuation and scaling cover every prescribed $\mu > 0$, with frequency allowed to vary.

The proof note also isolates the missing uniform virial control caused by an unbounded axial derivative of the multiplier. The functional obstruction alone is not the dynamical counterexample; the exact traveling solutions supply that stronger conclusion. The comparison is with public v3, not an asserted author- or journal-approved erratum.

Proof location: BNLS_Cylindrical_Claim_Adjudication_20261007, §§1–6; round7_analytic_adjudication.md.

How these should be counted

One core counterexample construction and its extensions address two identifiable external targets: the unconditional nonradial negative-energy extension, and Gou’s specific theorem. These are not two unrelated named conjectures. The later results below are additional mathematical answers; their existence must not erase this original accomplishment.

The counterexample family

Completed constructions, dynamical structure and exact scope

3. Zero momentum and finite variance do not restore universal collapse

In four dimensions, the project constructs a forward-global negative-energy solution with exactly zero total momentum, finite variance at every finite time, and a uniformly bounded H^{10} norm. It approaches two negative-energy waves moving in opposite directions. Its variance grows quadratically through separation, while its Sobolev norm stays bounded. This is a substantive strengthening of the counterexample, not a claim of two-sided global existence for the glued trajectory.

Proof location: BNLS_Zero_Momentum_and_Three_Dimensions_20261007, Theorem 1.1; round5_gluing_general.md and final review; round10_multisoliton_frontier.md.

4. The examples extend beyond isolated exact traveling solutions

The certified four-dimensional excited state has a fully determined symmetry kernel and Morse indices $n(L+) = 31$ and $n(L-) = 5$. A later certified radial complex spectral quartet completes the slow moving wave's stable and unstable real dimensions at 31. The corresponding waves are nonlinearly orbitally unstable. Strong stable families provide genuine nonstationary forward-global solutions converging exponentially to a wave, with exactly the same mass, momentum and negative energy. These finite-dimensional families are not an open set of stable initial data.

Proof location: BNLS_Dynamics_20261007, Theorems 2.1, 3.2 and 4.1; the subsequent zero-momentum manuscript, §6, upgrades the earlier dimension alternative to 31.

5. Robustness under specified changes of the equation

The early continuation supplies explicit small-velocity intervals and quantified small dispersion perturbations in four dimensions. Scaling extends the construction to arbitrary fixed real symmetric second-order dispersion matrices, with frequency free. Ellipsoidal quartic principal symbols and a neighborhood of the cubic power are also covered under the stated hypotheses. This does not establish the construction for every positive quartic symbol or every exponent.

Proof location: BNLS_Extensions_20261007, Theorems 2.1 and 3.1, Corollary 3.2 and Proposition 4.1.

Why later ground-state positivity is consistent

The early seed is an **excited stationary state**, not a sharp Gagliardo–Nirenberg optimizer. Its inverse dipole pairing is negative. The later positivity of $L-$ and of the response tensor T applies to sharp ground states. The two statements concern different profiles; neither overturns the other. The uniform mass gap in the later theory explains why these excited counterexamples cannot approach the scalar critical mass arbitrarily closely.

$$I = \langle \partial_1 Q, L_-^{-1} \partial_1 Q \rangle \le -\frac{24609689664617}{140499437500} < -175 \quad P_1(U_v) = 2Iv + O(v^3); \quad E(U_v) = Iv^2/4 + O(v^4) < 0.$$

The full nine-component four-dimensional finite-certificate runner was executed again on 8 October and all components passed. This reproduces the supplied implementations; it is not an independent external implementation or formal proof-assistant verification. The three-dimensional and gluing results retain their earlier full proof and internal review records.

Thresholds and coherent structures

Additional completed mathematical questions; not separately verified named historical conjectures

6. A strict mass gap for negative-energy traveling profiles

For real-even positive elliptic homogeneous dispersion of order $m > 1$, every negative-energy traveling profile has mass at least $Mc + \delta$, for some $\delta > 0$ independent of frequency and velocity. No radiality, variance, ground-state uniqueness or real-sector nondegeneracy is required. Specified positive subhomogeneous dispersion additions, defocusing additions and strictly positive-definite critical Hartree interactions are included. Common-frequency multicomponent profiles admit an internal-dimension-independent extension and near-threshold polarization rigidity.

This completely answers the profile question: the negative-energy traveling mechanism cannot occur arbitrarily close to Mc. It does not establish the analogous statement for every bounded time-dependent orbit, and the optimal gap is not computed.

Proof location: Mass_Gap_and_Momentum: thm:gap; R8: thm:r8-homogeneous-gap; R9: cone:gap; R10: thm:r10-hartree-gap; R11: thm:r11-vectorgap.

7. The limiting structure of least-mass negative-energy profiles

Assuming this profile class is nonempty, a normalized sequence minimizing its mass either converges strongly to a negative-energy minimizer or splits strongly into finitely many zero-energy exact profiles with common parameters. No diffuse remainder survives. Additional regular branch and Fredholm hypotheses give endpoint charge degeneracy and a neutral Jordan chain. The theory does not decide the minimum mass numerically, guarantee attainment in the negative-energy class, or prove time-dependent soliton resolution.

Proof location: R10: thm:r10threshold-compact; R11: thm:r11-end-attained, thm:r11-end-jordan and thm:r11-end-split.

8. Energy geometry and the cost of radiating momentum

For pure quartic negative-energy waves, a specified dilation plus linear-phase curve preserving mass and momentum has an exact positive energy difference from the wave. This is only a minimum along that curve, not orbital stability. Separately, general homogeneous negative-energy profiles force at least three negative directions in the full action Hessian. Smooth-branch response identities refine the index information.

For free radiation with prescribed mass μ and momentum π, the exact energy infimum is $a^{**}(\pi)/(2\mu^{(m-1)})$, where a^{**} is the convex envelope of the symbol. Schwartz data attain every strictly larger energy. Conditional applications give sharp invariant budgets for an already assumed coherent-core plus radiation decomposition; they do not construct that nonlinear decomposition.

Proof location: Mass_Gap_and_Momentum §5; R10: thm:r10-three-directions and thm:r10-response-index; R8: thm:r8-radiation-cost and thm:r8-decoupling.

Related predecessors include BLSS [3] for traveling-wave existence and symmetry, and Hamiltonian index theory [4]. A direct original-paper statement of each exact question above has not been established in the project's priority review.

How dispersion changes the answer

Variational conclusions and their dynamical consequences

9. A small same-order perturbation can destroy the profile gap

For b(ξ) = ε a(ξ)|ξ|^q/(1+|ξ|^q), one has 0 ≤ b ≤ εa, yet the modified equation admits negative-energy stationary waves at masses arbitrarily close above the original Mc. Thus positivity and small relative size alone do not preserve the previous profile gap. The dilation condition in the uniform theorem has real content. These are counterexamples for a modified equation, distinct from the early pure-BNLS constructions.

For the specified low-/high-frequency interpolation family, the project determines two variational mass thresholds, the intervening constrained energy, and an upper-endpoint classification. Near the lower threshold it derives leading energy and scale laws and selection on the optimizer manifold. These stationary scales are not temporal collapse rates; mixed-order stabilization itself has established predecessors.

Proof location: R9: thm:r9-bounded-boundary; R10: thm:r10-two-thresholds, thm:r10-endpoint and thm:r10-selection.

10. The exact energy of critical concentration under a bounded correction

Let k = a−d with d bounded and having a uniform directional high-frequency limit. The least limiting energy of a critical-mass concentrating sequence is computed by maximizing the limiting correction over the normalized sharp-optimizer manifold. Below this energy, mass-bounded energy sublevels are bounded in the energy space; above it, critical-mass sublevels contain unbounded sequences.

A conservative continuation flow gives all-time bounds for every datum in that region. A concrete quartic model in dimensions one through three includes the critical high-frequency mass endpoint at negative energy. Equality is model-dependent. This is a static concentration boundary, not a classification of dynamical blowup.

Proof location: R11: thm:r11-concentrating-sequence, cor:r11-sublevels, thm:r11-global-endpoint and cor:r11-rational-global.

11. Slow moving ground waves: constrained curvature and actual departure

For m > 2, sufficiently slow nonzero-speed action ground waves have a negative energy direction preserving mass and every component of momentum. Its leading curvature is $(2-m)v^{T}Tv$, with T positive. This variational conclusion needs neither real-sector nondegeneracy nor a differentiable profile branch.

Under the additional spectral hypotheses, the critical zero mode splits with $\lambda^2 = (m-2)v^{T}Tv/DQ + o(|v|^2)$. For pure quartic dispersion in dimensions one through three, the remaining substantive hypothesis is real-sector ground-state nondegeneracy. True nonlinear trajectories with exactly the wave's mass, momentum and energy leave its orbit, with leading exit time $\lambda^{-1}\log(1/\delta)$. The fate after departure is not classified.

Proof location: R12: thm:r12-constrained-curvature, cor:r12-all-ground-waves and thm:r12-nonlinear-instability; R11: thm:r11-spectral-pair.

Actual evolution and internal modes

What the later results add to the original counterexamples

12. Necessary transport and quantitative kinetic amplification

For zero-momentum negative-energy global motion, bounded kinetic averages force intrinsic mass separation at spatial scales proportional to the observation time. Under mass confinement, appropriate kinetic averages must instead grow. Related gradient-tail and center-drift conclusions quantify what globally bounded motion must do. These results are compatible with the earlier separating two-wave examples, which do not remain confined to one core. Residual estimates also force nonzero shape motion for specified near-threshold relative-periodic states; they do not exclude all breathers.

The R13 near-critical theorem gives a lower actual-time mean kinetic energy for a forward-global conservative zero-momentum solution without radiality, variance, symbol convexity or an a priori global kinetic bound. Write E = −e and d = (M/Mc)^(κ−1)−1, with κ = 1+m/N for local powers and κ = 2 for Hartree. For order m > 2,

$$\liminf_{T\to\infty} \frac{1}{T}\int_0^T A(t)\,dt \geq c\, e\, d^{-\alpha}, \quad \alpha = 1 + \frac{m}{2(m-1)}.$$

The theorem treats local critical powers and the specified positive-definite Hartree class. Near a real-sector nondegenerate quartic ground component, the exponent improves to 7/3 at zero momentum and to 2 at arbitrary momentum. Those two refinements remain conditional on nondegeneracy. The estimates still allow bounded motion at a sufficiently large kinetic level; they do not complete near-critical dynamical classification.

Proof location: Zero_Momentum_and_Three_Dimensions, Theorems 7.1–7.2; R11 transport theorems; R13: thm:r13-general-average, thm:r13-quadratic-drift and thm:r13-any-momentum.

13. Conserved occupation spectrum and global evolution

For common scalar dispersion with local total-density focusing, a Gaussian projection of the scalar sharp inequality produces an explicit coefficient depending on the full conserved internal occupation spectrum. The resulting condition gives uniform all-time energy-space bounds in mass intervals extending beyond the scalar Mc for any fixed non-rank-one occupation spectrum. At total mass exactly Mc, every evolution with unbounded energy norm must already lie in one fixed internal direction throughout its lifespan: U = ζu.

Examples are $M < \sqrt{2}\,Mc$ for two equal spatially orthogonal occupations in the two-dimensional quartic model, and $M < 4Mc/3$ in the two-dimensional cubic second-order model. These are sufficient intervals, not optimal collapse thresholds. Orthogonal polarization labels alone do not imply spatial L^2 orthogonality. The Gaussian bound is not asserted for Hartree interactions.

Proof location: R13: thm:r13-Gaussian and thm:r13-spectrumflow; the second-order example specializes r = 1 with the standard cubic energy-space flow.

Occupation conservation and operator inequalities have established predecessors [5,6]. The new coefficient’s priority and fixed-spectrum optimality remain separate questions. This work does not solve a Lieb–Thirring conjecture.

Scope, sources and the remaining work

A stable reading of the complete research line

What is complete, and what is still open in this project

The early constructions answer the original universal negative-energy collapse question negatively and contradict the specified cylindrical theorem. The later results answer additional profile, spectral, variational and dynamical questions under their written assumptions. Their number must not be converted into a count of independently resolved historical conjectures.

Still unresolved here: a general near-critical dynamical classification; the optimal mass threshold for negative-energy traveling waves; unrestricted ground-state real-sector nondegeneracy and uniqueness; an exact stable BNLS collapse law; the fate after slow-wave departure; general nonlinear soliton resolution; and optimal fixed-occupation thresholds. A global averaged inequality is not a finite-time blowup theorem. A static profile decomposition is not time-dependent soliton resolution.

No later note has been found withdrawing the early excited-state construction. Statements about excluding bounded negative-energy trajectories must be restricted to the specified near-critical class. They cannot be extended to all masses, since the earlier high-mass examples are already part of the project.

Primary literature: the two direct external targets

[1] T. Boulenger and E. Lenzmann, Blowup for biharmonic NLS, Ann. Sci. ÉNS 50(3) (2017), 503–544. §1.5 discusses removing radiality; its separate finite-time conjecture remains radial. Published full text.

[2] T. Gou, Blowup of cylindrically symmetric solutions for biharmonic NLS, public arXiv:2205.08167v3 (4 November 2024), Theorem 1.2. The corresponding work appeared in Proc. Edinburgh Math. Soc. 67(4) (2024), 1085–1098. The full public v3 is the exact text compared here. Public full text.

Selected predecessors for the subsequent theory

[3] Bugiera–Lenzmann–Schikorra–Sok, On symmetry of traveling solitary waves for dispersion generalized NLS, arXiv:1908.06771 (2019).

[4] Lin–Zeng, Instability, index theorem, and exponential trichotomy for linear Hamiltonian PDEs, Mem. AMS 275 (2022), no. 1347; arXiv:1703.04016.

[5] Hong–Kwon–Yoon, Global existence versus finite time blowup dichotomy for the system of nonlinear Schrödinger equations, arXiv:1712.06811.

[6] Dolbeault–Felmer–Loss–Paturel, Lieb–Thirring type inequalities and Gagliardo–Nirenberg inequalities for systems, J. Funct. Anal. 238 (2006), 193–220.

[7] Pausader–Shao, The mass-critical fourth-order Schrödinger equation in high dimensions, J. Hyperbolic Differ. Equ. 7 (2010), 651–705; arXiv:0906.1547.

[8] Côte–Le Coz, High-speed excited multi-solitons in nonlinear Schrödinger equations, J. Math. Pures Appl. 96 (2011), 135–166; arXiv:1007.3034.

Evidence standard. The source manuscripts, frozen data and implementation reviews support the internal proof record. The full research program has not undergone external independent reproduction, journal review or formal verification. No exhaustive first-priority claim is made. These limits do not erase existing proofs, but must accompany public claims.